Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$
$\theta : (a, b) \to \mathbb{R}^3$ such that

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$
$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$
$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$
There must be an acceleration keeping the curve on the surface

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction:

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$
$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$
There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p
Last time we observed:
For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$

Curves on surfaces

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

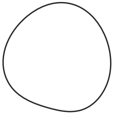
For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

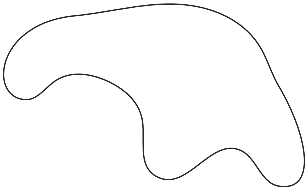
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

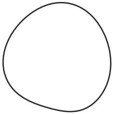
By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$





Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

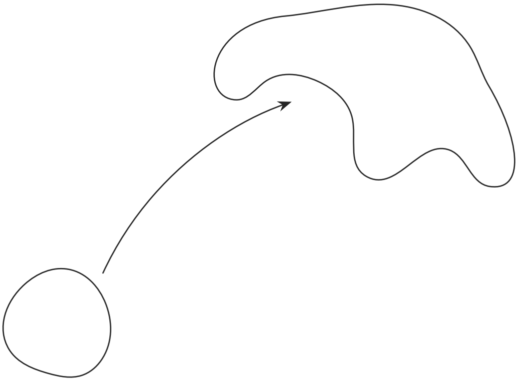
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

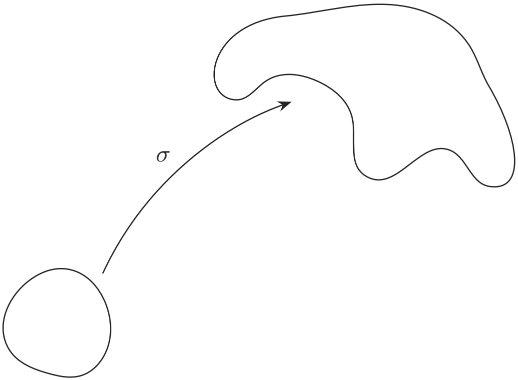
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

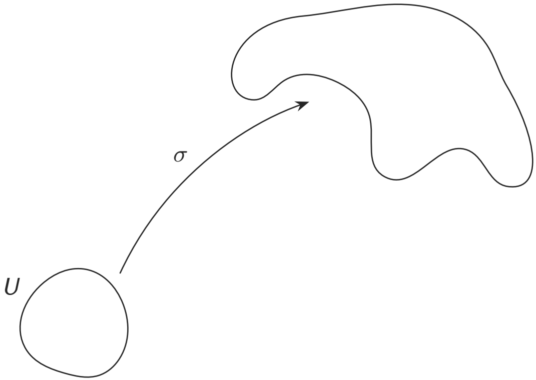
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

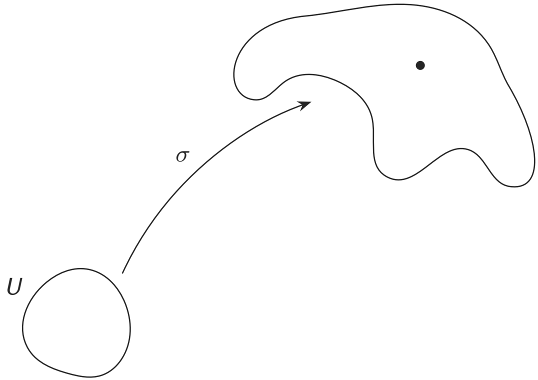
$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

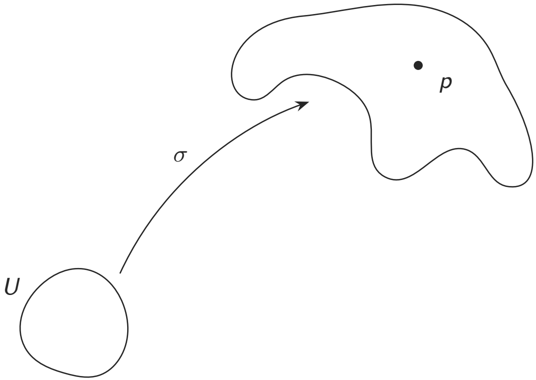
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

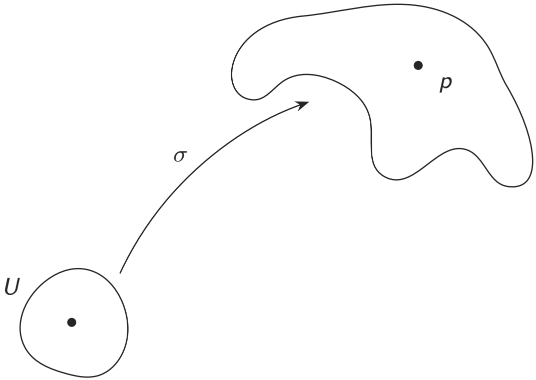
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

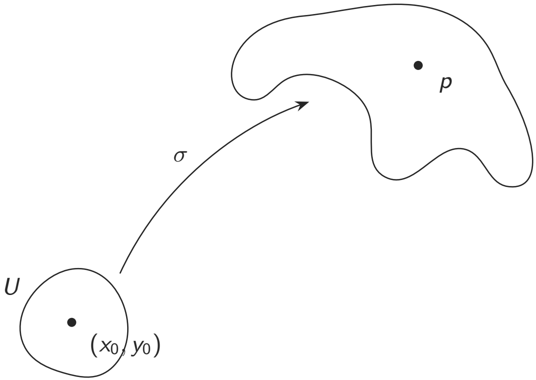
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

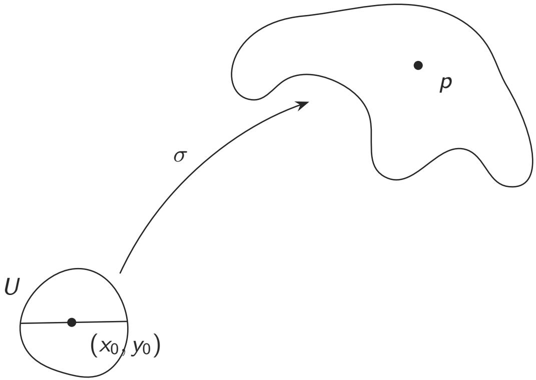
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

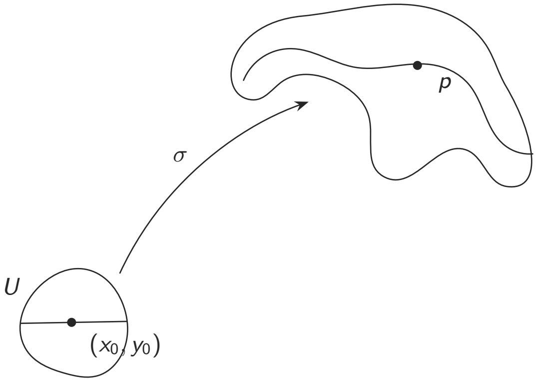
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

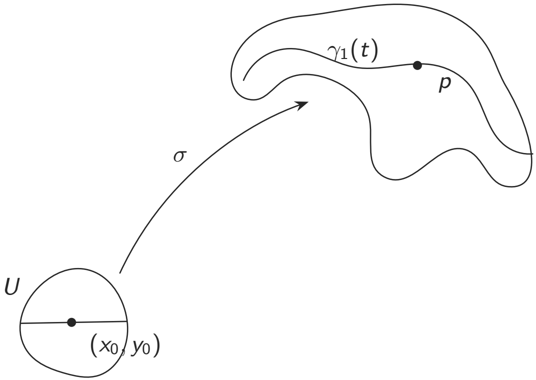
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

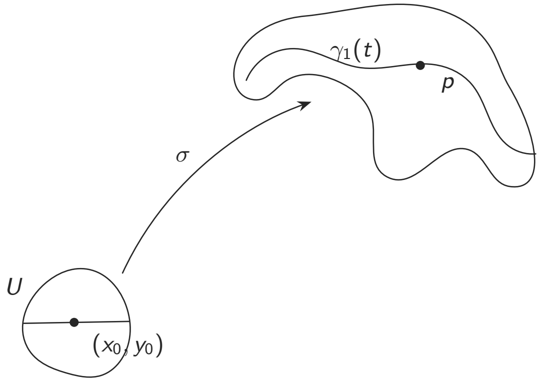
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) =$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

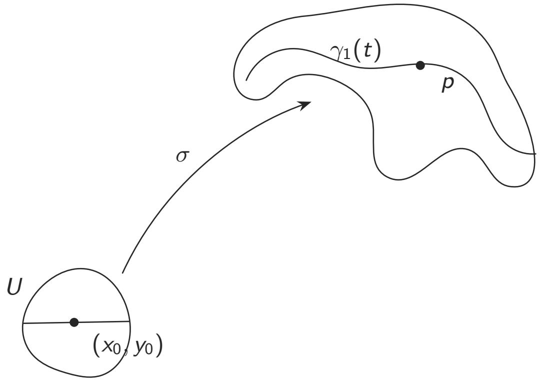
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$

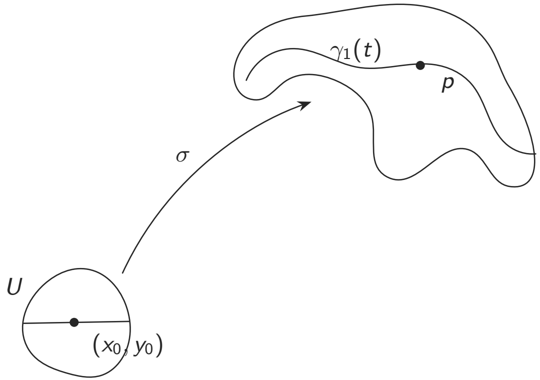


$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$

If $\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

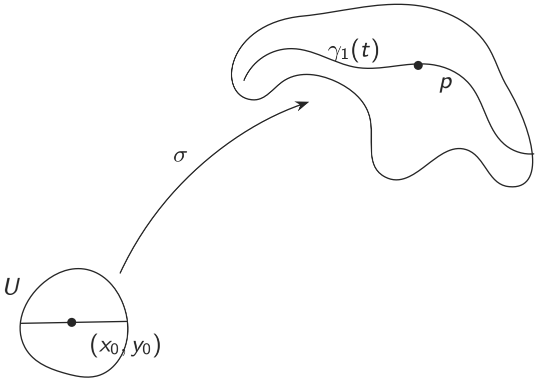
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

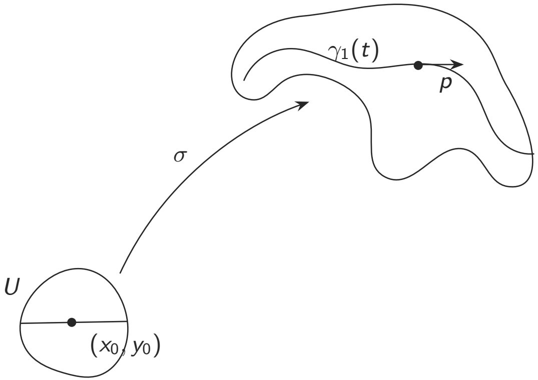
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

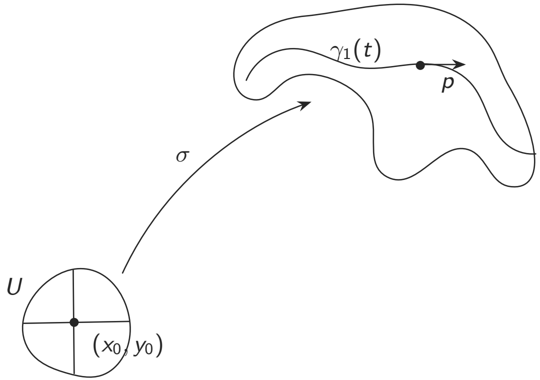
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

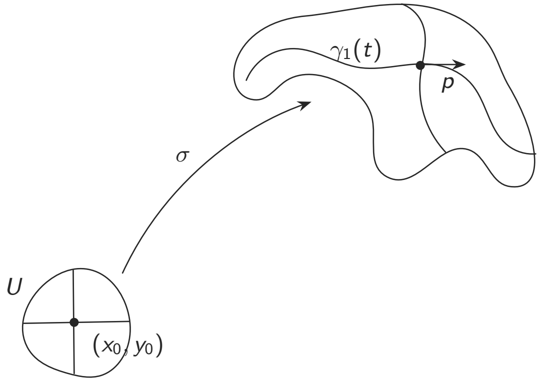
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

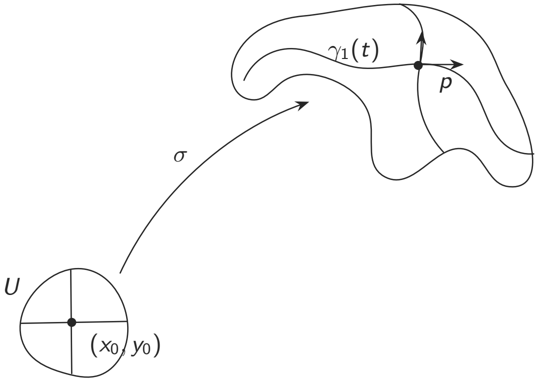
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

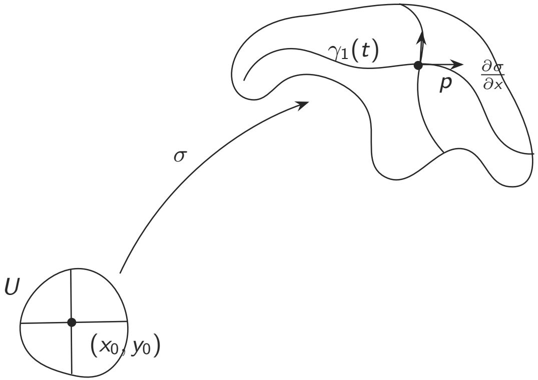
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

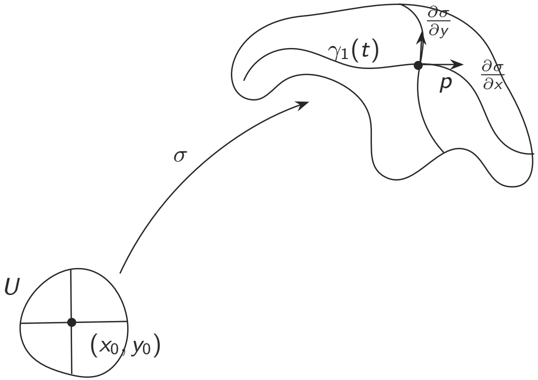
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

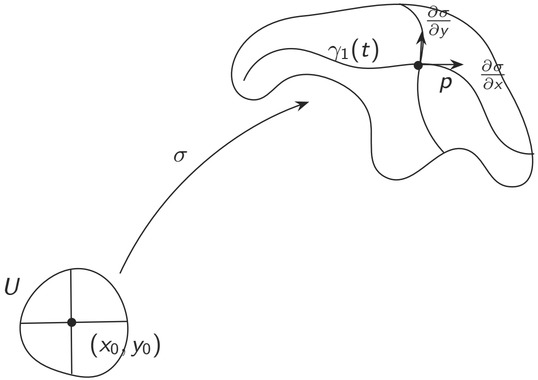
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

Curves on surfaces

Given $\sigma : U \to \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

$p \in \sigma(U)$

$\theta : (a, b) \to \mathbb{R}^3$ such that $\theta(t_0) = p$ and $\theta(a, b) \in \sigma(U)$

There must be an acceleration keeping the curve on the surface in the direction: perpendicular to the tangent plane of the surface at p

Last time we observed:

For any curve on the surface parametrized by γ , its velocity vector $\dot{\gamma}(t_0)$ is tangent to the surface

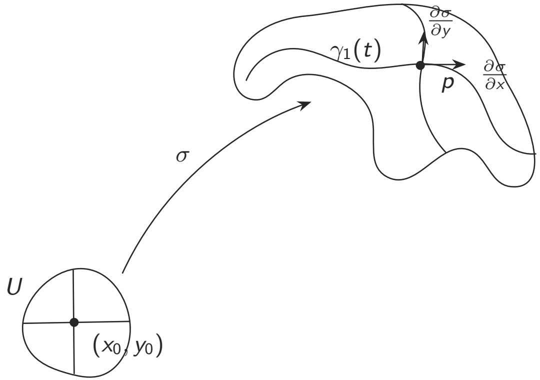
Therefore, we can define a tangent vector as the velocity vector of some parametrization of a curve that lies on the surface

By chain rule, we observed that $\dot{\gamma}(t_0)$ can be written in terms of two vectors

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$

If $\frac{\partial \sigma}{\partial x}$ *and* $\frac{\partial \sigma}{\partial y}$ *are to be tangent to the surface, then they should be velocity vectors of some curves. What are those two curves?*

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$

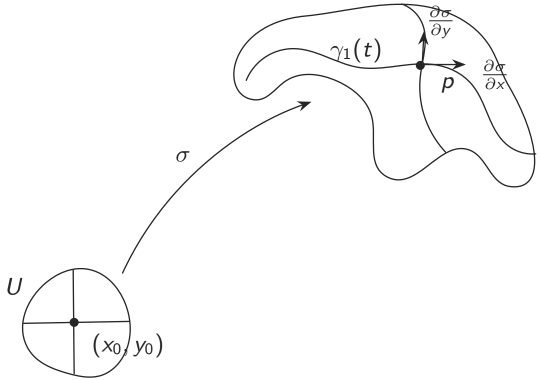


$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



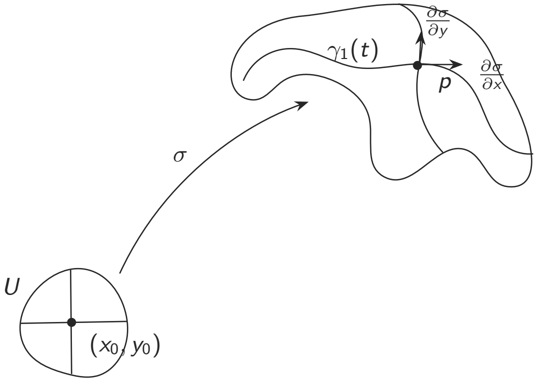
$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$

$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

$$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



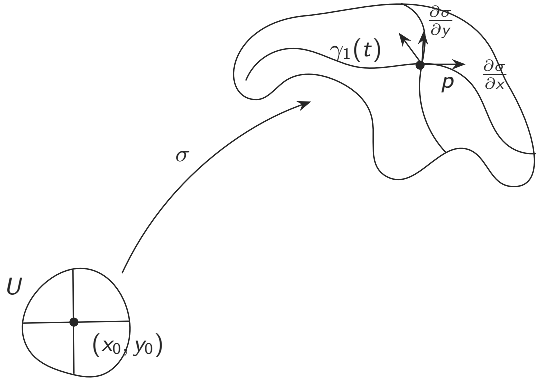
$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$

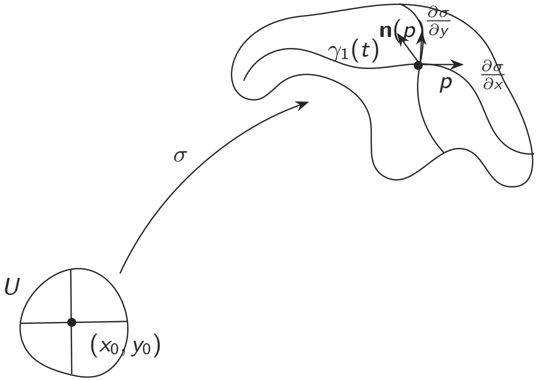
$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



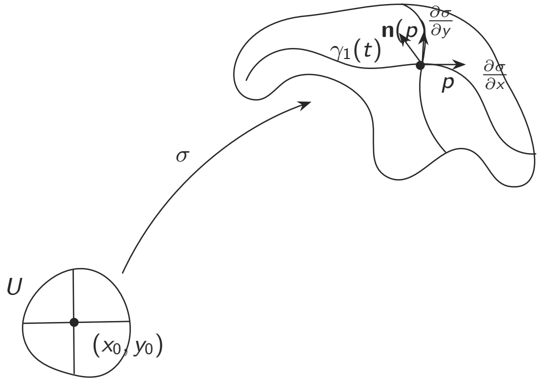
$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



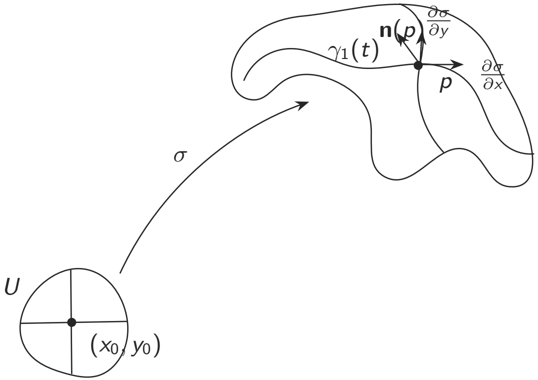
$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h,y_0)-\sigma(x_0,y_0)}{h} = \frac{\partial\sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$

$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$

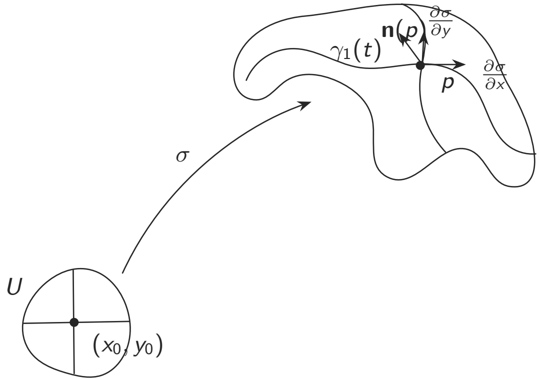
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$

$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

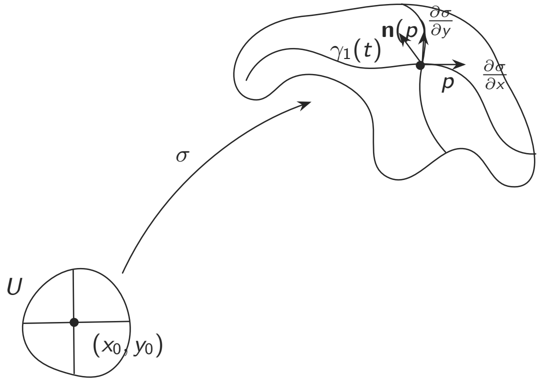
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

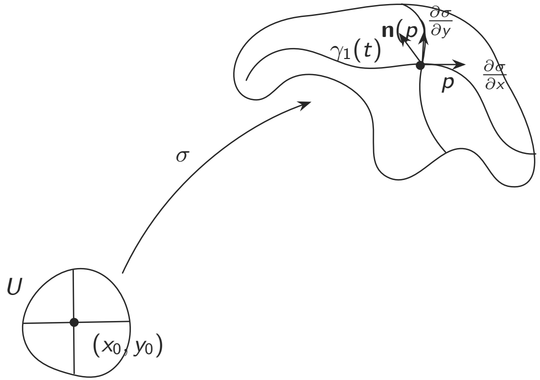
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

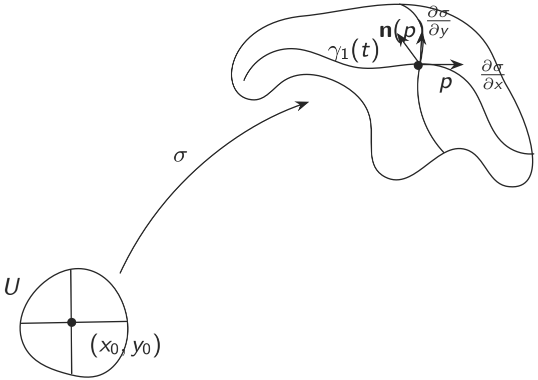
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



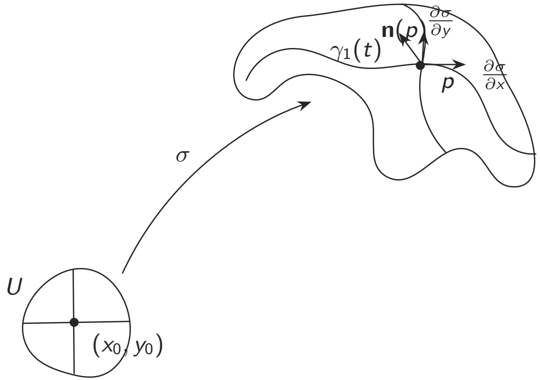
$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$
$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

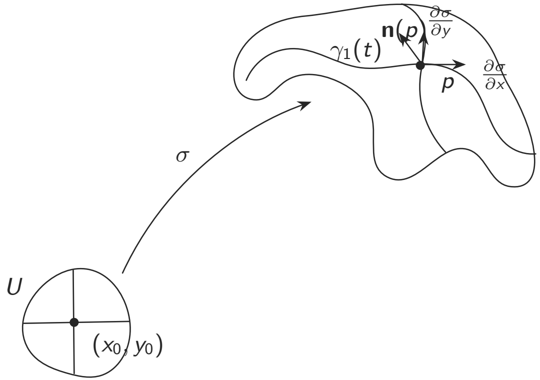
$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$
$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

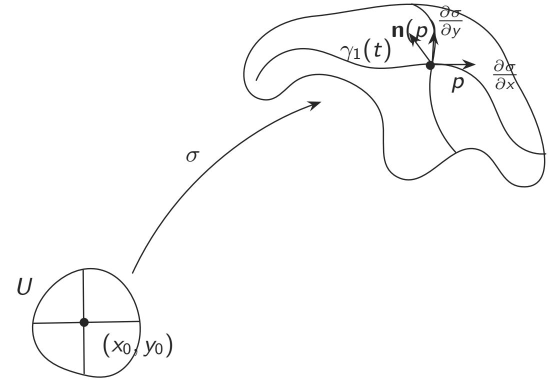
$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

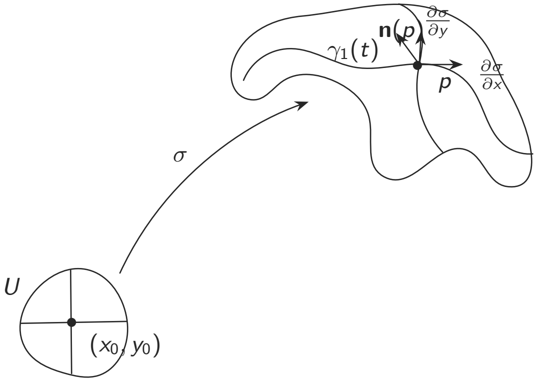
$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p

Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



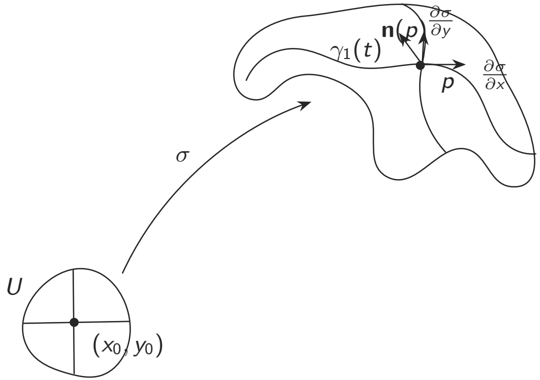
$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed

Since it lies on the surface,

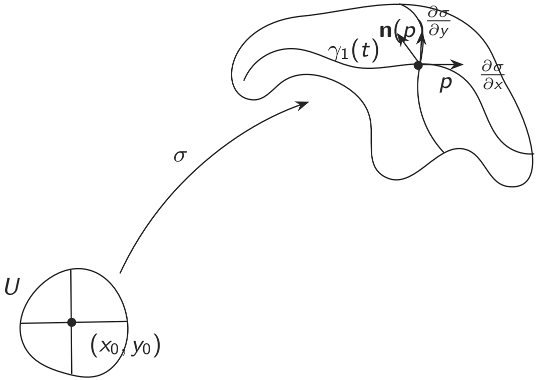
$\theta(t) = \sigma(x(t), y(t))$

$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$

Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$

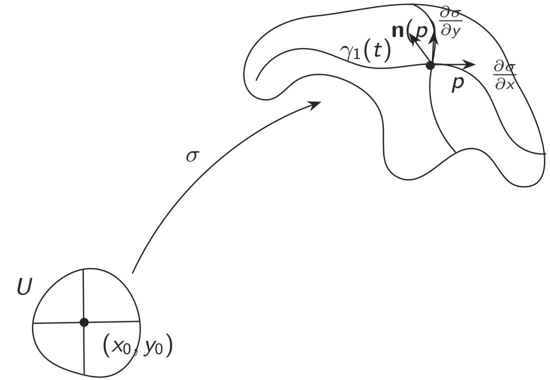
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



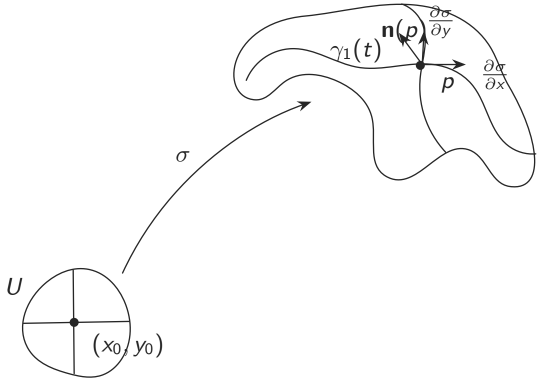
$\gamma_1'(0) = \lim_{h\to 0}\frac{\sigma(x_0+h,y_0)-\sigma(x_0,y_0)}{h} = \frac{\partial\sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$
$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



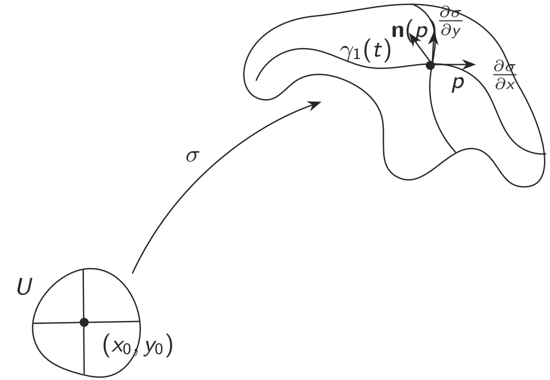
$\gamma_1'(0) = \lim_{h \to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial \sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial \sigma}{\partial y}$
$\frac{\partial \sigma}{\partial x}$ and $\frac{\partial \sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \neq 0$

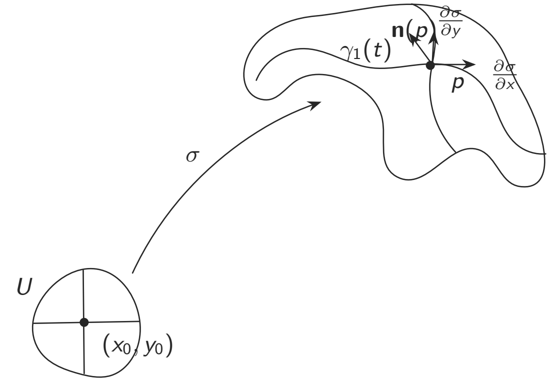
$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface

$\gamma_1(t) = \sigma(x_0 + t, y_0)$, where $\sigma(x_0, y_0) = p$



$\gamma_1'(0) = \lim_{h\to 0} \frac{\sigma(x_0+h, y_0) - \sigma(x_0, y_0)}{h} = \frac{\partial\sigma}{\partial x}$
Similarly, $\gamma_2(t) = \sigma(x_0, y_0 + t)$ is a curve whose velocity vector is $\frac{\partial\sigma}{\partial y}$
$\frac{\partial\sigma}{\partial x}$ and $\frac{\partial\sigma}{\partial y}$ will define a two dimensional space only if $\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \neq 0$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

○

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

○

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$,

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$,

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x\partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x\partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y\partial y}$,

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

-

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x\partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x\partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y\partial y}$, $\sigma_{yx} = \frac{\partial^2\sigma}{\partial y\partial x}$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}}{\|\frac{\partial\sigma}{\partial x}\times\frac{\partial\sigma}{\partial y}\|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

-

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x\partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x\partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y\partial y}$, $\sigma_{yx} = \frac{\partial^2\sigma}{\partial y\partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed

Since it lies on the surface,

$\theta(t) = \sigma(x(t), y(t))$

$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

The geodesic curvature at p:

$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$

Notice that $\theta''(t_0)$ is the sum of the two vectors

$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

◦

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$

$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2 \sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2 \sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$

$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) +$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed

Since it lies on the surface,

$\theta(t) = \sigma(x(t), y(t))$

$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

The geodesic curvature at p:

$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$

Notice that $\theta''(t_0)$ is the sum of the two vectors

$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$

$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x\partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x\partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y\partial y}$, $\sigma_{yx} = \frac{\partial^2\sigma}{\partial y\partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$

$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + \ x'(t)(\sigma_x(x(t), y(t)))' +$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed

Since it lies on the surface,

$\theta(t) = \sigma(x(t), y(t))$

$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

The geodesic curvature at p:

$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$

Notice that $\theta''(t_0)$ is the sum of the two vectors

$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$

$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2\sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$

$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' + y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

-

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2 \sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2 \sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$
$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' +$
$y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$
$= x''(t)\sigma_x(x(t), y(t)) +$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2 \sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2 \sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$
$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' +$
$y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$
$= x''(t)\sigma_x(x(t), y(t)) + x'(t)$

$\mathbf{n}(p) = \frac{\frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y}}{\| \frac{\partial\sigma}{\partial x} \times \frac{\partial\sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed

Since it lies on the surface,

$\theta(t) = \sigma(x(t), y(t))$

$\theta'(t) = x'(t)\frac{\partial\sigma}{\partial x} + y'(t)\frac{\partial\sigma}{\partial y}$

Consider the acceleration $\theta''(t)$

The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$

The geodesic curvature at p:

$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$

Notice that $\theta''(t_0)$ is the sum of the two vectors

$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

◦

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial\sigma}{\partial x}$ and $\sigma_y = \frac{\partial\sigma}{\partial y}$

$\sigma_{xy} = \frac{\partial^2\sigma}{\partial x\partial y}$, $\sigma_{xx} = \frac{\partial^2\sigma}{\partial x\partial x}$, $\sigma_{yy} = \frac{\partial^2\sigma}{\partial y\partial y}$, $\sigma_{yx} = \frac{\partial^2\sigma}{\partial y\partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$

$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' + y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$

$= x''(t)\sigma_x(x(t), y(t)) + x'(t)$
$(x'(t)\sigma_{xx}(x(t), y(t)) + y'(t)\sigma_{xy}(x(t), y(t))$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2 \sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2 \sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$
$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' + y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$
$= x''(t)\sigma_x(x(t), y(t)) + x'(t)$
$(x'(t)\sigma_{xx}(x(t), y(t)) + y'(t)\sigma_{xy}(x(t), y(t))$
$+ y''(t)\sigma_y(x(t), y(t)) +$

$\mathbf{n}(p) = \frac{\frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y}}{\| \frac{\partial \sigma}{\partial x} \times \frac{\partial \sigma}{\partial y} \|}$ is perpendicular to all the tangent vectors of the surface and is called the **normal**

Consider a curve on the surface parametrized by $\theta : (a, b) \to \mathbb{R}^3$ and passing through the point p and is unit speed
Since it lies on the surface,
$\theta(t) = \sigma(x(t), y(t))$
$\theta'(t) = x'(t)\frac{\partial \sigma}{\partial x} + y'(t)\frac{\partial \sigma}{\partial y}$

Consider the acceleration $\theta''(t)$
The normal curvature at p: $\theta''(t_0).\mathbf{n}(p)$
The geodesic curvature at p:
$\|\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p)).\mathbf{n}(p)\|$
Notice that $\theta''(t_0)$ is the sum of the two vectors
$(\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p) + (\theta''(t_0) - (\theta''(t_0).\mathbf{n}(p))\mathbf{n}(p))$

We can define the counterpart of a straight line in a plane for any surface , namely geodesics, as curves with geodesic curvature 0

-

If θ_1 and θ_2 are two unit speed parametrizations on the surface, both passing through p, their normal curvatures will be equal at p if the velocity vectors are the same

$\sigma_x = \frac{\partial \sigma}{\partial x}$ and $\sigma_y = \frac{\partial \sigma}{\partial y}$
$\sigma_{xy} = \frac{\partial^2 \sigma}{\partial x \partial y}$, $\sigma_{xx} = \frac{\partial^2 \sigma}{\partial x \partial x}$, $\sigma_{yy} = \frac{\partial^2 \sigma}{\partial y \partial y}$, $\sigma_{yx} = \frac{\partial^2 \sigma}{\partial y \partial x}$

$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$
$\theta''(t) = x''(t)\sigma_x(x(t), y(t)) + x'(t)(\sigma_x(x(t), y(t)))' + y''(t)\sigma_y(x(t), y(t)) + y'(t)(\sigma_y(x(t), y(t)))'$
$= x''(t)\sigma_x(x(t), y(t)) + x'(t)$
$(x'(t)\sigma_{xx}(x(t), y(t)) + y'(t)\sigma_{xy}(x(t), y(t))$
$+ y''(t)\sigma_y(x(t), y(t)) +$
$y'(t)(x'(t)\sigma_{yx}(x(t), y(t)) + y'(t)\sigma_{yy}(x(t), y(t)))$

-